

MFML AND MFDS

MFML and MFDS Team
WILPD - CSIS

BITS Pilani



innovate achieve lead

BITS Pilani

* ZC416 Mathematical Foundations for Machine Learning

Webinar-2

AGENDA

PROBLEMS ON

(1) Orthogonal Matrices

(2) Gram-Schmidt orthogonalization

(3) Characteristic polynomial

(4) Eigen values, Eigen vectors, and Spectral theorem

(5) Singular Value Decomposition

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

ORTHOGONAL MATRICES

A square matrix "A" is said to be **orthogonal** if its columns are orthonormal.

(**Orthonormal columns** means dot product of any two columns is zero and each column is of length 1.)

Equivalently, we can say $AA^T = A^TA = I$

Remark: Rows of the orthogonal matrices are also orthonormal.

Example $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ is an orthogonal matrix for all value of θ.

GRAM SCHMIDT PROCESS

- We perform Gaussian elimination on the matrix $\boldsymbol{A}^T\boldsymbol{A}$ where $\boldsymbol{A}$ contains the basis vectors as its columns.
- Upon Gaussian elimination on the augmented matrix we reduce $[\boldsymbol{A}^T\boldsymbol{A}|\boldsymbol{A}^T]$ to get $[\boldsymbol{U}|\boldsymbol{L}^{-1}\boldsymbol{A}^T]$ where $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{L}\boldsymbol{U}$.
- Now we shall show that $\boldsymbol{Q}^T = \boldsymbol{L}^{-1}\boldsymbol{A}^T$ is a matrix whose rows are orthogonal.
- Consider $\boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{L}^{-1}\boldsymbol{A}^T\boldsymbol{A}(\boldsymbol{L}^{-1})^T = \boldsymbol{U}(\boldsymbol{L}^{-1})^T =$ some upper triangular matrix
- But $\boldsymbol{Q}^T\boldsymbol{Q}$ is a symmetric matrix and can only be upper triangular if it is diagonal. Therefore, the columns of $\boldsymbol{Q}$ are orthogonal. They can be normalized to obtain an orthonormal basis.

Problem 1

1. Consider $\mathbb{R}^2$ and two basis vectors

$$\mathbf{v}_1 = (3,1)^T \quad \text{and} \quad \mathbf{v}_2 = (2,2)^T.$$

Put these vectors into columns of a matrix A such that

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}.$$

2. The next step is to perform Gaussian elimination on the following augmented matrix:

$$\left[A^T A \mid A^T\right] = \left[\begin{array}{cc|cc} 10 & 8 & 3 & 1 \\ 8 & 8 & 2 & 2 \end{array}\right].$$

3. On performing Gaussian elimination of this augmented matrix we end up with

$$\left[\begin{array}{cc|cc} 1 & 0.8 & 0.3 & 0.1 \\ 0 & 1 & -0.25 & 0.75 \end{array}\right].$$

Note that after the completion of Gaussian elimination the two rows on the right hand side are orthogonal. They form a basis for R2. We can normalize the vectors to get an orthonormal basis.

PROBLEM 2

Convert the given basis vectors into orthonormal basis of $\mathbb{R}^3$.

Given basis vectors are

$$a = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad c = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

SOLUTION:

Let A be a 3×3 matrix containing a, b and c as its columns. Then

$$A = \begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad \text{and} \quad A^T A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}$$

Now,

$$[A^T A | A^T] = \left[\begin{array}{ccc|ccc} 2 & 1 & -1 & 0 & -1 & 1 \\ 1 & 2 & 1 & 1 & 0 & 1 \\ -1 & 1 & 2 & -1 & 1 & 0 \end{array}\right]$$

Now, apply Gaussian elimination on $[A^TA \mid A^T]$, we have

$$\left[\begin{array}{ccc|ccc} 2 & 1 & -1 & 0 & -1 & 1 \\ 1 & 2 & 1 & 1 & 0 & 1 \\ -1 & -1 & 2 & -1 & 1 & 0 \end{array}\right] \xrightarrow[R_3 \to R_3 + \frac{R_1}{2}]{R_2 \to R_2 - \frac{R_1}{2}} \left[\begin{array}{ccc|ccc} 2 & 1 & -1 & 0 & -1 & 1 \\ 0 & \frac{3}{2} & \frac{3}{2} & 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{3}{2} & -1 & \frac{1}{2} & \frac{1}{2} \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 + \frac{R_2}{3}} \left[\begin{array}{ccc|ccc} 2 & 1 & -1 & 0 & -1 & 1 \\ 0 & \frac{3}{2} & \frac{3}{2} & 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 2 & -\frac{2}{3} & \frac{2}{3} & \frac{2}{3} \end{array}\right]$$

Note that the rows on the right hand side of the above matrix are **mutually orthogonal** and the squares of the lengths of these rows appear on the diagonal of the left side of the matrix. They form a basis for $\mathbb{R}^3$. We can normalize the vectors to get an **orthonormal basis**.

CHARACTERISTIC POLYNOMIAL:

For $\lambda \in \mathbb{R}$ and matrix $A \in \mathbb{R}^{n \times n}$,

we can define a polynomial, $\rho_A(\lambda) = \det(A - \lambda I)$.

$= c_0 + c_1\lambda + \cdots + c_{n-1}\lambda^{n-1} + (-1)^n c_n \lambda^n; c_0, c_1, \ldots, c_n \in \mathbb{R}.$

is the characteristics polynomial for the matrix A.

Problem: Let the matrix

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

The characteristic polynomial is

$$\det(A - \lambda I) = (1 - \lambda)^2 - 1.$$

EIGEN VALUES, EIGEN VECTORS, AND SPECTRAL THEOREM

EIGEN VALUE AND EIGEN VECTOR:

Let $A \in \mathbb{R}^{n \times n}$ be an $n \times n$ square matrix. $\lambda \in \mathbb{R}$ is called an **eigenvalue** of A and

$$x \in \mathbb{R}^n \setminus \{0\}$$

is the corresponding **eigenvector** of λ if $Ax = \lambda x$. This equation is called the eigenvalue equation.

SPECTRAL THEOREM:

If A is *nxn* symmetric matrix then there exists an orthonormal basis of the corresponding vector space V consisting of the eigenvectors of A, and each eigenvalue is real.

PROBLEM 3

Consider the given matrix

$$S = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix}.$$

Find an orthogonal matrix Q and a diagonal matrix D such that

$$S = QDQ^T.$$

SOLUTION

The characteristic equation is given by

$$\det(A - \lambda I) = 0.$$

Thus,

$$\begin{vmatrix} 2-\lambda & 0 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 0 & 2-\lambda \end{vmatrix} = 0.$$

Expanding the determinant, we obtain

$$(2-\lambda)^2(1-\lambda) - (1-\lambda) = 0$$
$$(1-\lambda)\left(\lambda^2 - 4\lambda + 4 - 1\right) = 0$$
$$(1-\lambda)(\lambda-1)(\lambda-3) = 0.$$

Therefore, the eigenvalues are

$$\lambda = 3, 1, 1.$$

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

For $\lambda = 3$, $\begin{bmatrix} -1 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & -1 \end{bmatrix} X = 0$

Row reducing, gives $\begin{bmatrix} -1 & 0 & 1 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix} X = 0$. Therefore,

$$-2x_2 = 0 \quad \Rightarrow \quad x_2 = 0.$$

Let $x_3 = t \quad \Rightarrow \quad x_1 = t$. Hence, $\therefore \quad v_1 = t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad t \neq 0.$

For $\lambda = 1$, $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} X = 0.$

Row reducing, gives $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} X = 0$. Let

$$x_2 = t_1, \qquad x_3 = t_2.$$

Then,

$$x_1 = -t_2.$$

Therefore,

$$\therefore \quad v_2 = \begin{bmatrix} -t_2 \\ t_1 \\ t_2 \end{bmatrix}, \qquad (t_1, t_2) \neq (0, 0).$$

$$\therefore \quad p_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \qquad p_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad p_3 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Clearly, p_1, p_2, p_3 are orthonormal.
Therefore,

$$P = \begin{bmatrix} p_1 & p_2 & p_3 \end{bmatrix}, \qquad P^{-1} = P^T = \begin{bmatrix} p_1^T \\ p_2^T \\ p_3^T \end{bmatrix}.$$

Thus,

$$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

SINGULAR VALUE DECOMPOSITION

SINGULAR VALUES:

The square root of eigenvalues of a symmetric matrix A^TA or AA^T are called singular values of the matrix A.

SINGULAR VECTORS:

The eigen vectors of A^TA corresponding to singular values of A are called singular vectors.

The singular vectors of A^TA are called right singular vectors and singular vectors of AA^T are called left singular vectors.

DEFINITION OF SVD:

Let A be any matrix of order $m \times n$, then this matrix can be decomposed into the product of three matrices given by

$$A_{m\times n} = U_{m\times m}\, \Sigma_{m\times n}\, V^T_{n\times n}$$

and it is called singular value decomposition of A.

Where U and V are called orthogonal matrices or unitary matrices and Σ is called the singular matrix.

WORKING RULE:

Step-1: Compute A^TA or AA^T.

Step-2: Find the eigenvalues of A^TA or AA^T.

Step-3: Compute the square roots of the non-zero eigenvalues obtained in the above step. These are called the singular values of A.

Step-4: Construct a singular matrix Σ of the same order as A, having singular values on the diagonal in decreasing order.

Step-5: Find the eigenvectors of AA^T or A^TA and construct an orthogonal matrix U or V consisting of orthonormal eigenvectors of A^TA.

Step-6: Construct the vectors $\sigma_i u_i = Av_i$ and hence obtain the orthogonal matrix U.

Step-7: Finally, $A = U\Sigma V^T$ gives the singular value decomposition (SVD) of matrix A.

PROBLEM 4

Find Singular value decomposition of the matrix $A = \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}$.

Solution: $A_{3\times 2} = U_{3\times 3}\Sigma_{3\times 2}V^T_{2\times 2}$

Step-1: Compute A^TA

$$\Rightarrow A^TA = \begin{bmatrix} -3 & 6 & 6 \\ 1 & -2 & -2 \end{bmatrix} \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix} = \begin{bmatrix} 81 & -27 \\ -27 & 9 \end{bmatrix}$$

Step-2: Find the eigenvalues of A^TA The characteristic equation (CE) of matrix A^TA is

$$\lambda^2 - \text{Trace}(A^TA)\lambda + |A^TA| = 0 \Rightarrow CE: \quad \lambda^2 - 90\lambda + 0 = 0$$

$$\Rightarrow \lambda(\lambda - 90) = 0$$

$\Rightarrow \lambda_1 = 90, \quad \lambda_2 = 0$. These are called eigenvalues of matrix A^TA.

Step-3: Next find the singular values of A: $\sigma_1 = \sqrt{\lambda_1} = \sqrt{90}$, $\sigma_2 = 0$

Step-4: Construct the singular matrix

Construct the singular matrix of size same as A and with its diagonal elements being singular values in decreasing order and all other entries are zeros (equivalent diagonal matrix).

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \sqrt{90} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Step-5: Find the eigenvectors of A^TA. The eigenvectors of A^TA are called right singular vectors of A with respect to each eigenvalue. For $\lambda = 90$

$$\Rightarrow A^TA - 90I = \begin{bmatrix} 81 & -27 \\ -27 & 9 \end{bmatrix} - 90\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -9 & -27 \\ -27 & -81 \end{bmatrix}$$

Hence from first row we get

$$-9x - 27y = 0$$

$\Rightarrow x = -3y = -3k; y = k$ is a free variable.
Therefore eigenvectors with respect to $\lambda = 90$ is

$$X_1 = k\begin{bmatrix} -3 \\ 1 \end{bmatrix} \Rightarrow v_1 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

For $\lambda = 0$

$$\Rightarrow A^T A - 0I = \begin{bmatrix} 81 & -27 \\ -27 & 9 \end{bmatrix} - 0 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 81 & -27 \\ -27 & 9 \end{bmatrix}$$

$$\Rightarrow y = k \text{ is a free variable.}$$

Hence from first row we get

$$81x - 27y = 0$$

$$\Rightarrow x = \frac{1}{3}y = \frac{1}{3}k$$

Therefore eigenvectors w.r.t. $\lambda = 0$ is $X_2 = \frac{k}{3}\begin{bmatrix} 1 \\ 3 \end{bmatrix} \Rightarrow v_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

Step-6: Write the orthogonal matrix.

Write the orthogonal matrix or right singular vectors matrix

$$V = [\hat{v}_1 \ \hat{v}_2]$$

$$V = \begin{bmatrix} \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \end{bmatrix}$$

Step-7: Find the left eigenvectors or left singular vectors

Find the left eigenvectors or left singular vectors $\hat{u}_i$ using

$$\hat{u}_i = \frac{Av_i}{\sigma_i}$$

For $i = 1$

$$u_1 = \frac{Av_i}{\sigma_i} = \frac{1}{\sqrt{90}} \begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix} \begin{bmatrix} \frac{3}{\sqrt{10}} \\ \frac{-1}{\sqrt{10}} \end{bmatrix} = \begin{bmatrix} \frac{-10}{\sqrt{900}} \\ \frac{20}{\sqrt{900}} \\ \frac{20}{\sqrt{900}} \end{bmatrix}$$

Since the second eigenvalue is zero,

$$\lambda_2 = \sigma_2 = 0$$

to find the other two vectors let us construct the orthogonal vectors using orthogonality conditions.

Let the next vector be $w = (x, y, z)$ then $w \cdot u_1 = 0$

$$\Rightarrow 10x - 20y - 20z = 0,$$

$\Rightarrow y = k_1$ and $z = k_2$
are free variables, hence

$$x = 2y + 2z = 2k_1 + 2k_2, \Rightarrow w = k_1 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

Hence the vectors orthogonal to u_1 are $w_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$, $w_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$

Let $u_2 = w_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \hat{u}_2 = \frac{u_2}{\|u_2\|} = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ 0 \end{bmatrix}$

$$u_3 = w_2 - \frac{\langle w_2, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} - \frac{4}{5} \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 2 \\ -4 \\ 5 \end{bmatrix}$$

$$\Rightarrow u_3 = \begin{bmatrix} 2 \\ -4 \\ 5 \end{bmatrix} \Rightarrow \hat{u}_3 = \frac{u_3}{\|u_3\|} = \begin{bmatrix} \frac{2}{\sqrt{45}} \\ -\frac{4}{\sqrt{45}} \\ \frac{5}{\sqrt{45}} \end{bmatrix}$$

Hence the orthogonal matrix U or matrix of left singular vectors is given by

$$U = \begin{bmatrix} \frac{10}{\sqrt{900}} & \frac{2}{\sqrt{5}} & \frac{2}{\sqrt{45}} \\ -\frac{20}{\sqrt{900}} & \frac{1}{\sqrt{5}} & -\frac{4}{\sqrt{45}} \\ -\frac{20}{\sqrt{900}} & 0 & \frac{5}{\sqrt{45}} \end{bmatrix}$$

Step-8:

Hence the singular value decomposition of given matrix A is

$$A = U\Sigma V^T$$

$$= \begin{bmatrix} \frac{10}{\sqrt{900}} & \frac{2}{\sqrt{5}} & \frac{2}{\sqrt{45}} \\ -\frac{20}{\sqrt{900}} & \frac{1}{\sqrt{5}} & -\frac{4}{\sqrt{45}} \\ -\frac{20}{\sqrt{900}} & 0 & \frac{5}{\sqrt{45}} \end{bmatrix} \begin{bmatrix} \sqrt{90} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -\frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \end{bmatrix}$$

PROBLEM 5

Given that the matrix

$$A = \begin{bmatrix} 1 & 0 & a \\ 0 & 2 & 0 \\ a & 0 & b \end{bmatrix}, \quad a, b \in \mathbb{R}$$

has 2 and -1 as its eigenvalues with algebraic multiplicity of 2 and 1 respectively. Further, the eigenvector corresponding to eigenvalue -1 is

$$\mathbf{e}_3 = \begin{bmatrix} 1 \\ 0 \\ -\sqrt{2} \end{bmatrix}$$

(a) Find the value of b.

(b) Write all the eigenvalues and their geometric multiplicity.

(c) By observation, and using the properties of a symmetric matrix, find the other two eigenvectors ($\mathbf{e}_1$ and $\mathbf{e}_2$) of A.

(d) Write the spectral decomposition of the matrix A.

(e) Find the value of a.

SOLUTION

(a) The eigenvalues of A are $2, 2, -1$. Hence,

$$1 + 2 + b = 2 + 2 - 1.$$

Thus, $b = 0$.

(b) Since A is symmetric, eigenvalue 2 has GM 2 and eigenvalue -1 has GM 1.

(c) The eigenvector corresponding to eigenvalue -1 is given by

$$\mathbf{e}_3 = \begin{bmatrix} 1 \\ 0 \\ -\sqrt{2} \end{bmatrix}.$$

SOLUTION

The other two eigenvectors correspond to eigenvalue 2 and both should lie in a plane orthogonal to $\mathbf{e}_3$. Thus, $\mathbf{e}_1$ and $\mathbf{e}_2$ are:

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} \sqrt{2} \\ 0 \\ 1 \end{bmatrix}$$

(d) The spectral decomposition of A is:

$$A = \begin{bmatrix} 0 & \frac{\sqrt{2}}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{3}} & -\frac{\sqrt{2}}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ \frac{\sqrt{2}}{\sqrt{3}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{\sqrt{2}}{\sqrt{3}} \end{bmatrix}.$$

SOLUTION

(e) As e_3 is an eigenvector corresponding to $\lambda = -1$, so we have

$$A.e_3 = -1.e_3$$

$$\Rightarrow \left(\begin{bmatrix} 1 & 0 & a \\ 0 & 2 & 0 \\ a & 0 & b \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right) .e_3 = 0$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & a \\ 0 & 3 & 0 \\ a & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 \\ 0 \\ -\sqrt{2} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$a = \sqrt{2}.$$

PROBLEM 6

Given that the Singular Value Decomposition of A is

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & e_{22} \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & \lambda_2 & \gamma_2 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{3\sqrt{2}} & -\frac{1}{3\sqrt{2}} & \frac{2\sqrt{2}}{3} \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

where

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$$

(a) Find the value of e_{22}.

(b) Find the value of λ_2.

SOLUTION

(a)

$$e_{22} = -\frac{1}{\sqrt{2}}.$$

(b) Utilizing the given decomposition of matrix A and comparing the corresponding elements on both sides we get ,

$$\frac{1}{\sqrt{2}}.\frac{5}{\sqrt{2}} + \frac{\lambda_2}{6} = 3$$

this gives, $\lambda_2 = 3$

PROBLEM 7

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

(a) Obtain the left-singular vectors of A.

(b) Obtain the right-singular vectors of A.

(c) Obtain the singular value matrix Σ. What is the spectral norm of A?

SOLUTION

(a)

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad A^T = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 5 & 2 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

SOLUTION

Eigenvalues and eigenvectors of AA^T:

$$\lambda_1 = \sigma_1^2 = 6, \qquad u_1' = \begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix}, \qquad \|u_1'\| = \sqrt{30}$$

$$\lambda_2 = \sigma_2^2 = 1, \qquad u_2' = \begin{bmatrix} 0 \\ 1 \\ -\frac{1}{2} \\ 1 \end{bmatrix}, \qquad \|u_2'\| = \frac{\sqrt{5}}{2}$$

$$\lambda_3 = \sigma_3^2 = 0, \qquad u_3' = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \qquad \|u_3'\| = \sqrt{6}$$

SOLUTION

Singular value matrix:

$$\Sigma = \begin{bmatrix} \sqrt{6} & 0 \\ 0 & \sqrt{1} \\ 0 & 0 \end{bmatrix}$$

Left singular vectors:

$$U = \left[u_1 = \frac{u_1'}{\|u_1'\|} \quad u_2 = \frac{u_2'}{\|u_2'\|} \quad u_3 = \frac{u_3'}{\|u_3'\|} \right]$$

SOLUTION

Right Singular Vectors:

$$v_1 = \frac{1}{\sigma_1} A^T u_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix} \frac{1}{\sqrt{30}} = \frac{1}{\sqrt{6}\sqrt{30}} \begin{bmatrix} 12 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$$

$$v_2 = \frac{1}{\sigma_2} A^T u_2 = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -\frac{1}{2} \\ 1 \end{bmatrix} \frac{2}{\sqrt{5}} = \begin{bmatrix} -\frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}$$

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

TAYLORS EXPANSION

PROBLEM 8

If the function is defined as

$$f(x) = x^2 + \cos x + \sin x,$$

find the second degree Taylor's polynomial of f at $x = \frac{\pi}{2}$.

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

SOLUTION

$$f(x) = x^2 + \cos x + \sin x, f\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4} + 1,$$

$$f'(x) = 2x - \sin x + \cos x, f'\left(\frac{\pi}{2}\right) = \pi - 1$$

$$f''(x) = 2 - \cos x - \sin x, f''\left(\frac{\pi}{2}\right) = 2 - 0 - 1 = 1.$$

Therefore, by Taylor's theorem about $x = \frac{\pi}{2}$,

$$T_2(x) = f\left(\frac{\pi}{2}\right) + \left(x - \frac{\pi}{2}\right) f'\left(\frac{\pi}{2}\right) + \frac{\left(x - \frac{\pi}{2}\right)^2}{2!} f''\left(\frac{\pi}{2}\right)$$

$$= \left(\frac{\pi^2}{4} + 1\right) + (\pi - 1)\left(x - \frac{\pi}{2}\right) + \frac{\left(x - \frac{\pi}{2}\right)^2}{2}.$$

Hessian Matrix

The **Hessian matrix** is a square matrix containing the **second-order partial derivatives** of a multivariable function.
For a function of two variables, $f(x, y)$, the Hessian matrix is defined as

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

More generally, for a function

$$f : \mathbb{R}^n \to \mathbb{R},$$

the Hessian matrix is denoted by

$$H_f(\mathbf{x}) = \nabla^2 f(\mathbf{x})$$

where the (i, j)-th entry of the Hessian matrix is

$$(H_f)_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$$

Thus, the general Hessian matrix can be written as

$$H_f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

PROBLEM 9

Consider the energy-loss function at a point (x, y) given by

$$f(x, y) = [x, y]Q[x, y]^T + a[x, y]^T,$$

where

$$Q = \begin{bmatrix} 2 & 1 \\ 2 & 4 \end{bmatrix} \quad \text{and} \quad a = [1, 5].$$

(i) Compute the gradient of f and find the co-ordinate of the point P at which the gradient of f is zero.

(ii) Evaluate the Hessian matrix at that point P.

SOLUTION

(i) $$f(x,y) = [x,y]\begin{bmatrix}2 & 1\\ 2 & 4\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix} + [1,5]\begin{bmatrix}x\\ y\end{bmatrix}$$

$$\therefore \quad \nabla f = [x,y]\left(\begin{bmatrix}2 & 1\\ 2 & 4\end{bmatrix} + \begin{bmatrix}2 & 2\\ 1 & 4\end{bmatrix}\right) + [1,5]$$

$$= [x,y]\begin{bmatrix}4 & 3\\ 3 & 8\end{bmatrix} + [1,5] = [4x+3y+1,\ 3x+8y+5]$$

For the gradient to be zero, $4x+3y+1=0, \quad 3x+8y+5=0.$

$$\text{This gives} \quad x = \frac{7}{23}, y = -\frac{17}{23}$$

Hence, the point is $P = \left(\frac{7}{23}, -\frac{17}{23}\right)$.

(ii)

$$H = \begin{bmatrix}f_{xx} & f_{xy}\\ f_{yx} & f_{yy}\end{bmatrix} = \begin{bmatrix}4 & 3\\ 3 & 8\end{bmatrix}$$

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Problem 10

Consider the loss function

$$f(x, y) = e^x \cos y + e^y \sin x.$$

(i) Compute the Hessian matrix $H_f(x, y)$ of the function f.

(ii) Evaluate the Hessian matrix at $(0, 0)$ and prove or disprove that $H_f(0, 0)$ is symmetric positive definite.

SOLUTION

The first-order partial derivatives are

$$f_x = e^x \cos y + e^y \cos x, f_y = -e^x \sin y + e^y \sin x.$$

The second-order partial derivatives are

$$f_{xx} = e^x \cos y - e^y \sin x, f_{xy} = -e^x \sin y + e^y \cos x,$$

$$f_{yx} = -e^x \sin y + e^y \cos x, f_{yy} = -e^x \cos y + e^y \sin x.$$

Therefore, the Hessian matrix is

$$H_f(x, y) = \begin{bmatrix} e^x \cos y - e^y \sin x & -e^x \sin y + e^y \cos x \\ -e^x \sin y + e^y \cos x & -e^x \cos y + e^y \sin x \end{bmatrix}$$

(ii)

$$H_f(0,0) = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

The matrix is symmetric because

$$H_f(0,0)^T = H_f(0,0).$$

The leading principal minor is $1 > 0$, but

$$\det H_f(0,0) = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2 < 0.$$

Therefore, the matrix is indefinite, and hence it is not symmetric positive definite.



Thank you